

Arnica Sarker

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🌐 GitHub 🔗 LinkedIn 🌐 Portfolio

Computer Science and Engineering undergraduate (CGPA 3.96/4.00) seeking admission to a Master's/Ph.D. program in Computer Science, with research interests in Machine Learning, Deep Learning, Natural Language Processing, and Explainable AI (XAI). Second-author publication at an IEEE international conference; undergraduate thesis applying temporal deep learning and explainable AI methods to clinical time-series data.

RESEARCH INTERESTS

Machine Learning • Deep Learning • Natural Language Processing • Explainable AI (XAI) • Applied AI in Healthcare

EDUCATION

B.Sc. in Computer Science & Engineering (CSE)

University of Asia Pacific (UAP), Dhaka, Bangladesh

Expected December 2026

- Registration ID: 22201015
- CGPA: 3.96/4.00

Higher Secondary Certificate (H.S.C.)

Cumilla Victoria Government College, Cumilla, Bangladesh

2020

- GPA: 5.00/5.00

Secondary School Certificate (S.S.C.)

Nawab Faizunnesa Government Girls' High School, Cumilla, Bangladesh

2018

- GPA: 5.00/5.00
- Board Scholarship
- Government Scholarship, Junior Level (Class 8)

RESEARCH EXPERIENCE

Undergraduate Thesis

Jan. 2026 – Present

Department of CSE, University of Asia Pacific

Supervisor: Assistant Professor Molla Rashied Hussein

- “Early Sepsis Prediction Using ICU Time-Series Data with Explainable AI and Resource-Adaptive AI” — developing a temporal deep learning model for early sepsis prediction from ICU time-series data, incorporating explainable AI methods to support clinical interpretability.
- Developing a temporal deep learning model for early sepsis prediction using the MIMIC-IV ICU database.
- Integrating Explainable AI (SHAP/LIME) to improve the interpretability of clinical predictions.
- Investigating resource-efficient AI models for deployment in healthcare environments.

Optical Sensing Research

Sep 2025 – Apr. 2026

Department of CSE, University of Asia Pacific

Performance Analysis of High Sensitivity WDM-FBG Based Optical Sensing Applications

- Contributed to a study analyzing the performance of high-sensitivity WDM-FBG based optical sensing applications, resulting in a Second-author IEEE conference publication.
- Proposed a WDM-FBG-based optical sensing architecture for simultaneous temperature, stress, and strain monitoring.
- Demonstrated high sensing sensitivities (0.75 dB/°C, 0.20 dB/MPa, and 0.075 dB/μ ϵ) while maintaining stable spectral integrity in WDM simulations.

PUBLICATIONS

- [1] M. T. Rahman, **A. Sarker**, and A. Naeem, “Performance Analysis of High Sensitivity WDM-FBG Based Optical Sensing Applications,” *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*, IEEE, pp. 1–6, 2026.

HONORS AND AWARDS

- Vice Chancellor's Honor Award (six semesters), University of Asia Pacific
- Dean's Honor Award (one semester) for outstanding academic performance, University of Asia Pacific

- 1st Prize, Robo Expo 2.0, organized by Robotics Club, UAP-CSE
- 1st Runner-up, Regional Mathematical Olympiad 2016, organized by Math Olympiad Bangladesh
- 1st Runner-up, Regional Mathematical Olympiad 2013, organized by Math Olympiad Bangladesh

TECHNICAL SKILLS

Programming Languages:	Python, C, C++, Java
Machine Learning & AI:	Scikit-learn, TensorFlow, Keras, PyTorch, NLP, Hugging Face Transformers
Web Development:	Django, HTML, CSS, JavaScript
Explainable AI (XAI):	SHAP, LIME, Feature Importance Analysis
Data Analysis & Visualization:	NumPy, Pandas, SciPy, Matplotlib, Seaborn
Model Evaluation:	Cross-Validation, Hyperparameter Tuning, ROC-AUC, Precision, Recall, F1-Score, Confusion Matrix
Development Tools:	kaggle notebook, Google Colab, Git, GitHub, Visual Studio Code, Pycharm, Codeblocks
Databases:	MySQL
Data Communication & Networks:	OptiSystem
Circuit Design Tools:	Tinkercad, Circuit Designer
Prototype Design Tools:	AutoCAD

RELEVANT COURSEWORK

Machine Learning and Deep Learning • Data Structures and Algorithms I & II • Computer Architecture and Organization • System Analysis and Design • Object-Oriented Programming I & II • Structured Programming • Microprocessors and Microcontrollers • Data Communication and Computer Networks • Computer and Cyber Security • Blockchain • Competitive Programming • Robotics • Operating Systems • VLSI • Numerical and Mathematical Analysis for Engineers

TECHNICAL PROJECTS

Explainable Early Sepsis Prediction Using Temporal Deep Learning and XAI

- Developed an explainable early sepsis prediction system using **Python, TensorFlow, LSTM, SHAP, and Hugging Face** for interpretable ICU time-series prediction.

Medical Insurance Cost Prediction

- Built a regression model using **Python, Scikit-learn, Pandas, and Matplotlib** to predict medical insurance costs from patient attributes.

CleanTrack+

- Developed a full-stack waste management system using **HTML, CSS, JavaScript, and MySQL** with authentication, complaint management, and an admin dashboard.

HandyNest

- Built a home-service management platform using **Java, and MySQL** for service booking and provider management.

QuantaDOT

- Designed a Bluetooth-controlled educational robot using **Arduino, Embedded C++, and HC-05 Bluetooth** for interactive STEM learning.

IoT-Based Smart Assistive Walking Solution

- Developed an IoT-enabled smart walking aid using **Arduino, GSM, ultrasonic, gas, and fall detection sensors** for obstacle detection and emergency alerts.

REFERENCES

Molla Rashied Hussein

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Dr. Muhammad Towfiqur Rahman

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